Hints: Abstract algebra – 2. Finite rings and division rings

Homework

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. Put $u = (1-ab)^{-1}$ and simply multiply out $(1-ba)(1+bua) = 1 - ba + b(1-ab)ua$, and the same on the other side.

Hint 2. $(ba)^{k+1} = b(ab)^k a$ gives the nilpotency statement. For the example take E_{12} and E_{21} in $M_2(\mathbb{R})$: their sum squares to I.

Problem 2

Note. The zero element must be counted among the zero divisors, i.e. zero divisors here means non-units. Otherwise $\mathbb{Z}/4$ is a counterexample: it has 2 nilpotents and only 1 non-zero zero divisor.

(a)

Hint 1. If $x^a = 0$ and $y^b = 0$, expand $(x+y)^{a+b}$ by the binomial theorem (the ring is commutative). Products are easy.

(b)

Hint 1. In a finite ring the zero divisors are exactly the non-units. Show that the set D of non-units is a union of cosets $d + N$ of the additive subgroup N of nilpotents.

Hint 2. If $d + n$ were a unit u with $n \in N$, then $d = u - n = u(1 - u^{-1}n)$ would be a unit, because $1-$(nilpotent) is a unit. So $|N|$ divides $|D|$.

Problem 3

Note. The statement is false if "characteristic zero" only means $n \cdot 1 \neq 0$ for all n. In $\mathbb{Z} \times \mathbb{F}_2$ the idempotents $e = f = (0,1)$, $g = 0$ satisfy $e + f + g = 0$. The hints assume that R has no additive torsion ($nx = 0$ with $n \neq 0$ implies $x = 0$), which is the intended meaning.

Hint 1. Square $e + f = -g$: you get $ef + fe = -2(e+f)$. Multiply this identity by e on the left, and separately on the right.

Hint 2. The two products give $efe = -2e - 3ef = -2e - 3fe$, hence $ef = fe$. Then $2ef = -2e - 2f$; multiplying by e and by f gives $4ef = -2e = -2f$.

Hint 3. So $e = f$, and by symmetry $e = f = g$. Then $3e = 0$ gives $e = 0$.

Problem 4

Hint 1. The element $e = x^{n-1}$ is an idempotent with $ex = xe = x$. In the corner ring eRe the element x is invertible (with inverse x^{n-2}, or e if $n = 2$).

Hint 2. For $m \geqslant 2$, try $u_m = x^{m-1} + (1-e)$. Then $u_m x = x^m$, because $(1-e)x = 0$.

Hint 3. Its inverse is $x^k + (1-e)$, where $k \geqslant 1$ is chosen so that $m - 1 + k$ is a multiple of $n-1$; then $x^{m-1+k} = e$. The cross terms vanish because $x(1-e) = (1-e)x = 0$. For $m = 1$ take $u_1 = 1$.

Problem 5

Hint 1. Let S be the sum of all elements. Pairing x with $-x$ shows $2S = 0$. If S is invertible, then $2 = 2S \cdot S^{-1} = 0$, so the characteristic is 2.

Hint 2. Then $(A, +) \cong (\mathbb{Z}/2)^k$. The sum of all vectors of $\mathbb{F}_2^k$ is 0 for $k \geqslant 2$. Hence $|A| = 2$, $A \cong \mathbb{Z}/2$, and there $S = 1$ is invertible.

Problem 6

Hint 1. $(1+1)^2 \in \{0, 1\}$, so $4 = 0$ or $3 = 0$. In the case $3 = 0$, test $(a+1)^2 = a^2 + 2a + 1$ against $a^2 \in \{0, 1\}$; you should get $A = \{0, 1, -1\} = \mathbb{F}_3$.

Hint 2. Now let $4 = 0$ and $N = \{a : a^2 = 0\}$. Show N is a two-sided ideal. (i) If $(na)^2 = 1$ or $(an)^2 = 1$ for some $n \in N$, then n has a one-sided inverse, which is impossible when $n^2 = 0$. (ii) If $a, b \in N$ and $ab + ba = (a + b)^2 = 1$, then $(ab)^2 = ab$ with $ab \in N$, so $ab = 0$ and likewise $ba = 0$.

Hint 3. Every $a \notin N$ has $a^2 = 1$, so A/N is a division ring in which $x^2 = 1$ for all $x \neq 0$. Hence $A/N \in \{\mathbb{F}_2, \mathbb{F}_3\}$, and $\mathbb{F}_3$ was the first case.

Hint 4. Answer: $A \cong \mathbb{F}_3$, or A has a two-sided ideal N with $A/N \cong \mathbb{F}_2$ such that $n^2 = 0$ and $2n = 0$ for every $n \in N$. (Check the converse from $(1+n)^2 = 1+2n$ and $(n+m)^2 = 0$.) Examples are $\mathbb{F}_2$, $\mathbb{Z}/4$, $\mathbb{F}_2[X]/(X^2)$ and $\mathbb{F}_2[X, Y]/(X, Y)^2$.

Problem 7

Hint 1. Squares are central, so $(x+1)^2 - x^2 - 1 = 2x$ is central for every x. If $\operatorname{char} K \neq 2$, divide by 2.

Hint 2. In characteristic 2, suppose $xy \neq yx$. Then $c = xy + yx = (x + y)^2 - x^2 - y^2$ is central and non-zero. Compute $(xy)^2$ using $xyx = cx + x^2y$.

Hint 3. You get $(xy)^2 = c\,xy + x^2y^2$. Hence $c\,xy$, and so xy, is central. In particular x commutes with xy, which gives $xy = yx$, a contradiction.

Problem 8

Hint 1. If A is a field with $q = 2n + 1$ elements, then $x^q = x$, and for $k < q$ the polynomial $X^k - X$ has fewer than q roots. So $\min M = q$.

Hint 2. Conversely, $x^k = x$ for all x implies there are no non-zero nilpotents. In a reduced ring idempotents are central: $(er(1-e))^2 = 0$. So A splits as $eA \times (1-e)A$. Iterate until no non-trivial idempotents are left.

Hint 3. A finite factor with no non-trivial idempotents, satisfying $x^k = x$, is a division ring (x^{k-1} is an idempotent, hence 0 or 1), hence a field by Wedderburn. So $A \cong \mathbb{F}_{q_1} \times \cdots \times \mathbb{F}_{q_r}$ with all q_i odd.

Hint 4. If $r \geqslant 2$, then $k = \operatorname{lcm}(q_i - 1) + 1 \leqslant \prod(q_i - 1) + 1 < \prod q_i = 2n + 1$ lies in M. So $\min M = 2n + 1$ forces $r = 1$.

Problem 9

Hint 1. In a field, $x^d = 1$ has at most d solutions. Let $\psi(d)$ be the number of elements of order d in $\mathbb{F}_q^*$. If $\psi(d) > 0$, these elements all lie in one cyclic group of order d, so $\psi(d) \leqslant \varphi(d)$.

Hint 2. $\sum_{d|q-1} \psi(d) = q-1 = \sum_{d|q-1} \varphi(d)$ forces $\psi(d) = \varphi(d)$ for all d, in particular $\psi(q-1) > 0$. In a cyclic group of order $q - 1$, $x^d = 1$ has exactly d solutions when $d \mid q - 1$.

Hint 3. For the sum: if $q > 2$, take a generator $g \neq 1$. Then $\sum_{x \neq 0} x = \sum_{i=0}^{q-2} g^i = \frac{g^{q-1}-1}{g-1} = 0$.

Problem 10

(a)

Hint 1. If $u = ef$ is a unit with e, f idempotent, then e has a right inverse fu^{-1}. An idempotent with a right inverse equals 1: $e = e \cdot e(fu^{-1}) = e(fu^{-1}) = 1$. Similarly $f = 1$.

(b)

Hint 1. By (a), $-1 = 1$. If $a^2 = 0$, then $1 + a$ is a unit, so $1 + a = 1$. Thus A has no non-zero elements of square zero.

Hint 2. Then every idempotent is central: $(er(1-e))^2 = 0$ and $((1-e)re)^2 = 0$. So for $x = ef$ with e, f commuting idempotents, $x^2 = e^2 f^2 = ef = x$.